

Customer Management Module

End User Documentation

Assignment Submission — Web Application Development

Date: April 10, 2026

Submission Details

GitHub Repository	https://github.com/your-username/customer-management-module	
Live Demo URL	https://customer-mgmt-module.onrender.com	
Hosting Platform	Render.com (Free Tier)	
Tech Stack	Node.js · Express · MySQL · HTML/CSS/JS · JWT	

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1. Introduction

The **Customer Management Module** is a web-based application designed to help administrators efficiently manage customer data — including personal information, account settings, and account status. It allows the organisation to keep an up-to-date record of all customers and provide support when needed.

This document serves as the complete reference guide for end users, covering all four functionalities, the database structure, REST API endpoints, deployment steps, and troubleshooting guidance.

1.1 Purpose

- Allow customers to self-register and manage their own profile.
- Provide administrators with a central dashboard to view and manage all customers.
- Enable admins to update customer contact information and account status.
- Implement a soft-delete mechanism so that deactivated accounts are never permanently lost.

1.2 Scope

This module focuses on customer account management. It does not include product management, order processing, or payment handling (covered in separate modules).

2. System Requirements

2.1 Server

Runtime	Node.js v18+ (or compatible backend runtime)
Database	MySQL 8.0+ or MariaDB 10.6+
Memory	Minimum 512 MB RAM
OS	Ubuntu 20.04+ / Windows Server 2019+ / macOS 12+

2.2 Client

Browser	Chrome 100+, Firefox 100+, Edge 100+, Safari 15+
JavaScript	Must be enabled in browser
Screen Width	Minimum 1024 px

2.3 NPM Dependencies

- `express` ^4.18 — HTTP server
- `mysql2` ^3.6 — MySQL driver
- `bcryptjs` ^2.4 — Password hashing
- `jsonwebtoken` ^9 — JWT authentication
- `cors` ^2.8 — Cross-origin middleware
- `dotenv` ^16 — Environment config

3. Getting Started

3.1 Live Application

The module is hosted at:

```
https://customer-mgmt-module.onrender.com
```

■ ■ Free-tier servers sleep after 15 min inactivity — first load may take up to 60 seconds.

3.2 Demo Credentials

Admin Email	admin@demo.com
Admin Password	Admin@123
Customer Email	rahul@example.com
Password	Customer@123

3.3 Local Setup

- Clone the repo:

```
git clone https://github.com/your-username/customer-management-module.git
```

- Install dependencies:

```
npm install
```

- Run the database schema:

```
mysql -u root -p < db/schema.sql
```

- Copy and edit the env file:

```
cp .env.example .env # Fill in DB_HOST, DB_USER, DB_PASS, DB_NAME, JWT_SECRET
```

- Start the server:

```
npm run dev # Runs at http://localhost:3000
```

4. Module Functionalities

4.1 Add a New Customer

There are two ways to add a customer to the system:

Option A — Customer Self-Registration (Frontend)

- Navigate to the application URL and click **Register here**.
- Fill in First Name, Last Name, Email, Phone, and Password.

- Click **Create Account**. On success you are redirected to login.
- The new account is created with **status = TRUE** (active) by default.

Option B — Admin Manually Adds Customer (Dashboard)

- Login as Admin and open the **Customer Dashboard**.
- Click the **+ Add Customer** button (top-right).
- Fill in all fields in the modal form and click **Save**.
- The customer record appears instantly in the dashboard table.

4.2 Customer Dashboard

The Customer Dashboard is the central view for Admins to monitor all customer accounts. It displays a paginated table with powerful search and filter options.

- View all customers with columns: ID, Full Name, Email, Phone, Status, Created Date.
- Search by name, email, or phone using the live search bar.
- Filter by account status: **All / Active / Inactive**.
- Stats row at the top shows Total, Active, and Inactive counts.
- Click **View** on any row to see the full customer detail modal.

4.3 Update Customer Details

Admins can modify any customer's contact information and account status. Customers can update only their own profile (name, email, phone) — they cannot change their own status.

Admin Steps:

- Open the dashboard and click **Edit** next to the target customer.
- Update any field in the Edit modal: First Name, Last Name, Email, Phone, Status.
- Click **Save**. The *updated_at* timestamp is refreshed automatically.

Customer Steps:

- Login as customer and click **My Profile** in the sidebar.
- Click **Edit My Details**. Update name, email, or phone.
- Click **Save**.

4.4 Delete / Deactivate Account

Customer accounts are never permanently deleted. The module implements **soft delete**: when an admin deactivates an account, the *status* field is set to **FALSE** and the customer can no longer log in. All data is preserved for audit purposes.

- Click **Deactivate** next to the customer in the dashboard.
- Confirm the prompt. The badge changes from *Active* → *Inactive*.
- To restore the account, click **Reactivate** — sets *status* = *TRUE*.
- Inactive customers cannot log in and see a "credentials invalid" error.

5. Database Design

The module uses a **users** table as its primary store, plus an **admins** table for admin accounts.

Table: users

Column	Datatype	Constraint	Description
user_id	INT	PK, AUTO_INCREMENT	Unique identifier for each customer
first_name	VARCHAR(100)	NOT NULL	Customer's first name
last_name	VARCHAR(100)	NOT NULL	Customer's last name
email	VARCHAR(100)	NOT NULL, UNIQUE	Customer's email address
phone	VARCHAR(15)	NULL	Customer's phone number
password	VARCHAR(255)	NOT NULL	Bcrypt-hashed password
created_at	DATETIME	DEFAULT NOW()	Timestamp when account was created
updated_at	DATETIME	ON UPDATE NOW()	Timestamp when details were last updated
status	BOOLEAN	DEFAULT TRUE	TRUE = active, FALSE = inactive (soft delete)

DDL — Create Table

```
CREATE TABLE users ( user_id INT NOT NULL AUTO_INCREMENT, first_name VARCHAR(100) NOT NULL, last_name VARCHAR(100) NOT NULL, email VARCHAR(100) NOT NULL UNIQUE, phone VARCHAR(15), password VARCHAR(255) NOT NULL, created_at DATETIME DEFAULT CURRENT_TIMESTAMP, updated_at DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP, status BOOLEAN DEFAULT TRUE, PRIMARY KEY (user_id) );
```

6. API Reference

Method	Endpoint	Description
POST	/api/auth/admin/login	Admin login — returns JWT token
POST	/api/auth/register	Customer self-registration
POST	/api/auth/login	Customer login — returns JWT token
GET	/api/customers	List customers (admin: all; customer: own)
GET	/api/customers/:id	Get single customer by ID
POST	/api/customers	Admin adds customer manually (admin only)

PUT	/api/customers/:id	Update customer details
DELETE	/api/customers/:id	Soft deactivate account (admin only)
PATCH	/api/customers/:id/reactivate	Reactivate account (admin only)

7. Error Messages & Troubleshooting

Error	Cause	Fix
400 – Validation Failed	Missing required fields	Ensure all required fields are provided in the request body.
401 – Unauthorised	Missing/expired JWT	Re-login to get a fresh token.
403 – Forbidden	Customer doing an admin action	Only admins can deactivate/reactivate accounts.
404 – Not Found	Invalid customer ID	Verify the ID exists in the dashboard.
409 – Email Already Exists	Duplicate email on register	Use a different email address.
409 – Already Inactive	Deactivating an inactive account	Account is already deactivated — no action needed.
500 – Server Error	DB connection issue / bug	Check .env credentials and MySQL service status.
Login fails for customer	Account may be inactive	Admin must reactivate the account first.

8. Deployment Guide (Render.com)

Step 1 — Push to GitHub

- Create a new GitHub repo and push all source files.

Step 2 — Create Render Web Service

- Go to render.com → New + → Web Service → Connect GitHub repo.
- Build Command: npm install | Start Command: node server.js

Step 3 — Add Environment Variables

- In Render dashboard → Environment tab, add: DB_HOST, DB_USER, DB_PASS, DB_NAME, JWT_SECRET, PORT.

Step 4 — Free MySQL (Aiven.io)

- Sign up at aiven.io → Create a free MySQL service.
- Copy the host, user, password, and database name into Render env vars.
- Run db/schema.sql on your Aiven MySQL instance to create tables and seed data.

Step 5 — Deploy

- Click Create Web Service. Render builds and deploys automatically.
- Your live URL will be something like: <https://customer-mgmt-module.onrender.com>

■ ■ Free tier sleeps after 15 min inactivity. First request may take 30–60 seconds.

9. GitHub Repository Guide

Complete source code hosted at:

<https://github.com/your-username/customer-management-module>

Repository Structure

```
customer-management-module/ ■■■ server.js # Express app entry point ■■■ routes/ ■  
■■■ auth.js # POST /api/auth/* routes ■ ■■■ customers.js # CRUD /api/customers/*  
routes ■■■ middleware/ ■ ■■■ auth.js # JWT verification + adminOnly guard ■■■ db/  
■ ■■■ connection.js # mysql2 connection pool ■ ■■■ schema.sql # DDL + seed data  
■■■ public/ ■ ■■■ index.html # SPA frontend ■ ■■■ css/style.css ■ ■■■ js/app.js  
■■■ .env.example ■■■ package.json ■■■ README.md
```

10. Frequently Asked Questions

Q: Can a deactivated customer's data be recovered?

A: Yes. Soft delete means only status = FALSE is set. Admins can reactivate at any time via the Reactivate button.

Q: Can two customers share the same email?

A: No. The email column has a UNIQUE constraint. Attempting to register a duplicate email returns HTTP 409.

Q: Can a customer change their own account status?

A: No. The PUT /api/customers/:id endpoint ignores the status field for non-admin users.

Q: Is the password stored in plain text?

A: No. All passwords are hashed using bcrypt with a salt rounds value of 10 before being stored.

Q: What happens if an admin tries to deactivate themselves?

A: Admins are stored in a separate admins table and cannot be deactivated through this module.

Q: Is there pagination on the dashboard?

A: Yes. GET /api/customers supports ?page= and ?limit= (default 15). The response includes total_pages.
